

9(a)	<i>any eight from:</i> 1 DNA denatured / ssDNA produced ; 2 by heating to, 94 / 95 °C ; 3 add primer (DNA) ; 4 complementary base pairing (with sample DNA) ; 5 at 55–65 °C ; 6 <i>ref. to</i> annealing ; 7 DNA polymerase builds new strands / AW ; 8 by adding free nucleotides ; 9 at 70–75 °C ; 10 <i>ref. to Taq</i> polymerase thermostable ; 11 does not need replacing ; 12 new strand denatured and process repeated ; 13 process is automated ;	8
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Question	Answer	Marks
9(b)	<i>any seven from:</i> 1 can produce large(r) quantities ; 2 use bacterial host / hamster cell host / insect larva cell ; 3 product exactly the same as human protein ; 4 (as) product has same amino acid sequence 5 no immune response ; 6 no side effects ; 7 no risk of transfer of disease ; 8 easier to obtain purified product ;	7